



FEEDBACK

YOUR SUGGESTIONS

EXPECT BETTER ARTICLES

In the current state of engineering education, practical electronics is missing. Startups and SMEs need practical knowledge, which they cannot get through books. There should be a medium (magazine or portal) through which this can be disseminated.

Circuits in EFY are more hobby-oriented and, off late, I am seeing articles written by professional writers and academicians that have no real educational values.

I hardly read EFY for more than 20 minutes as most articles are too general (or collated from the Internet), which may be good enough for a novice. For instance, I saw a series of articles on laser weapons, which does not have much value even for a hobbyist. Articles should contain material that a common engineer cannot access easily.

I have been working with many startups and I see most of them fail because they have no idea how to execute and scale up. We should aim to fill this white space.

Srinivasa Moorthy

Through email

EFY. Thanks for the candid and valuable feedback! Continuous improvement is our strategy and we will surely take your inputs seriously. EFY is for the entire electronics fraternity and so our goal is to keep them updated with the latest information related to electronics industry. Information includes technology trends, industrial and consumer electronics, innovations, smart world, Make in India, startups, designs and much more.

Designs and DIY sections are for professional engineers, students and hobbyists. DIY section is just one part of EFY contents. The section may include small- or medium-level projects for hobbyists, experimenters and circuit designers. However, since even a small circuit plays an important role in a bigger and complex project, so in true sense, all

circuits are equally important depending on the application. For example, water-level controller is a small circuit and yet a very useful one! A creative person can tweak the circuit and make a very successful commercial product using the same.

Regarding the articles on laser weapons, those are in continuation of another series of related articles covered in earlier issues by the same author, which you might have missed. The articles covered include basic principles to actual practical applications, which is very informative for electronics circuit designers.

SMART WORLD: THOUGHT LEADERS

This is regarding 'We are Moving Towards Smartphone Cameras That Capture The Same Quality Photographs as DSLRs' article that appeared on <https://smart.electronicsforu.com>

I am in disagreement here. I am using iPhone 8 Plus and D40x Nikon DSLR camera with manual settings and Nikkor lenses. If I capture the same subject using both devices, pictures taken with the DSLR come out much better with better colour depth and vibrancy.

Pratyush Paul

❑ Absolutely, Pratyush, you have spoken my heart. Smartphones can be an option at the most, but can never be a replacement for serious photography features that are present in DSLRs.

Sabya Ghosh

THERMAL IMAGING CAMERAS

This is regarding 'Thermal Imaging for Industrial Testing' article published in April issue. Please share some details regarding various thermal imaging cameras that can be fitted in a drone for inspection of solar farms and power industries.

A.J. Prakash

EFY. FLIR has many heat vision cameras of various resolutions, thermal frame rates, lenses, fields of view, colour scales and measurement accuracies, such as FLIR Vue Pro R and FLIR Duo Pro R. The company teamed up with DJI to create drone-based FLIR thermal imaging kits—specifically for fire fighters, first responders and building inspectors—which could be adapted for many other sectors.

In March 2018, DJI introduced their latest innovation in thermal visual technology with the release of Zenmuse XT2. This is an integrated thermal sensor having a visual sensor. It uses FLIR Tau 2 thermal core along with a DJI 4k visual camera mounted on DJI's Zenmuse gimbal. This latest thermal camera is housed in a weather-resistant casing.

Then, there is Workswell WIRIS camera, which is designed for drones and covers large areas such as power plants. During flight, it can monitor live images to produce a fully-radiometric record or sequence images of the whole flight. Records can be then analysed by Workswell CorePlayer software, which has advanced measuring functions and the option to easily and quickly produce reports. Its alarm mode indicates high temperature if it exceeds set limits. High digital zoom up to 16x is available; in the case of a thermal camera, up to 14x (with 640 × 512 resolution) is available. GPS navigation is an additional system accessory, fully manageable using a standard RC controller during flight. System parameters can be set without any stopover. You can change emissivity settings, pallets, temperature range and more. The camera is fully calibrated and comes with a calibration certificate.

There are many other manufacturers for drone cameras such as ABJ Drones, Flytron (V3 Micro thermal imaging camera), Yuneec (CGOET thermal imaging camera), Thermoteknix (Micro-CAM3 thermal vision camera). You may check their prices and features online.